Haosong Peng

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EDUCATION

The Hong Kong University of Science and Technology — **Hong Kong SAR, China**

Ph.D. Student in Division of Integrative Systems and Design — *Feb. 2025 – Now*

- **Research Interests**: Embodied Intelligence, 3D Vision, and Edge Intelligence | **Supervisor**: Prof.Wenchao Xu

Beijing Insitiute of Technology — **Beijing, China**

Master of Science in Engineering — *Sep. 2022 – Jun. 2025*

- **Research Interests**: Video Analytics System, Multimedia, Edge Computing | **Supervisor**: Prof. Yufeng Zhan
- Member of Institute of Intelligent Information Processing and Control, School of Automation | **Team Lead**: Prof. Yuanqing Xia

Beijing University of Chemical Technology — **Beijing, China**

Bachelor of Science in Engineering — *Sep. 2018 – Jun. 2022*

- **Major:** Automation **GPA**: 4.07/4.33 (Rank 1/153)

PUBLICATIONS & PREPRINTS

* = *Equal Contributions*

- TCP4D: A Unified Tool-Center-Point Tracking and Generation Model for Robot Manipulation

  Haosong Peng*, Xiangyang Liu*, Zhengshen Zhang, Yunfeng Fan, Tianyu Qi, Xin Xia, Yufeng Zhan and Wenchao Xu.

  Public preview, 2026.

- YANAMI: A Large-Scale Multimodal Dataset and Benchmark for Multiturn Stateful Interaction

  Shengyu Wang*, Tengyue Nong*, Junming Liu*, **Haosong Peng***, Siyu Li*, Ruiqi Lei, Fushuo Huo, Yuhao Pan, Yunfeng Fan, Ding Wang, Nan Cheng, Haozhao Wang and Wenchao Xu.

  arXiv preprint arXiv:2611.05280, 2026.

- World-to-Wrist: Task-Conditioned Future Wrist Modeling for Fine-Grained Robot Manipulation

  Yuhao Pan*, **Haosong Peng***, Zhengshen Zhang, Zhengyang Yan, Yalun Dai, Fushuo Huo, Chujie Wang, Tianyu Qi, Xiucheng Wang, Nan Cheng and Wenchao Xu.

  arXiv preprint arXiv:2608.05369, 2026.

- SpatialBench: Is your spatial foundation model an all-round player?

  Haosong Peng*, Hao Li*, Jiaqi Chen*, Yuhao Pan*, Runmao Yao, Yalun Dai, Fushuo Huo, Fangzhou Hong, Zhaoxi Chen, Haozhao Wang, Dingwen Zhang, Ziwei Liu and Wenchao Xu.

  NeurIPS 2026, Evaluations and Datasets Track.

- OmniVGGT: Omni-Modality Driven Visual Geometry Grounded Transformer

  Haosong Peng*, Hao Li*, Yalun Dai, Yushi Lan, Yihang Luo, Tianyu Qi, Zhengshen Zhang, Yufeng Zhan, Junfei Zhang, Wenchao Xu and Ziwei Liu.

  CVPR 2026, Highlight.

- DGTR: Distributed Gaussian Turbo-Reconstruction for Sparse-View Vast Scenes

  Hao Li*, Yuanyuan Gao*, **Haosong Peng***, Chenming Wu, Weicai Ye, Yufeng Zhan, Chen Zhao, Dingwen Zhang, Jingdong Wang and Junwei Han.

  ICRA 2025.

- Arena: A Patch-of-Interest ViT Inference Acceleration System for Edge-Assisted Video Analytics

  Haosong Peng*, Wei Feng*, Hao Li, Yufeng Zhan, Ren Jin and Yuanqing Xia.

  IEEE Transactions on Computers, 2026.

- Radiant: Efficient Timely Large-Scale Scene Analytics Based on Hierarchical Framework

  Haosong Peng*, Tianyu Qi*, Yufeng Zhan, Hao Li, Yalun Dai and Yuanqing Xia.

  IEEE Transactions on Services Computing, 2026.

- Tangram: High-resolution Video Analytics on Serverless Platform with SLO-aware Batching

Haosong Peng, Yufeng Zhan, Peng Li and Yuanqing Xia.

ICDCS 2024.

- Egret: Reinforcement Mechanism for Sequential Computation Offloading in Edge Computing

  Haosong Peng, Yufeng Zhan, Dihua Zhai, Xiaopu Zhang and Yuanqing Xia.

  IEEE Transactions on Services Computing, 2024.

SKILLS & INTERESTS

Technical: Python & Pytorch, C, R, Matlab & Simulink, Latex, Linux & Bash, Docker, OpenCV, MS Office;

Languages: English (CET-6: 538, IELTS: 7.0), Mandarin (Native)

ACADEMIC SERVICE

CVPR (2027), WWW (2027), ACM Multimedia (2024, 2026), Computer Networks (2026) — Reviewer

AWARDS & HONORS

- 2024 Excellent Student of Beijing Institute of Technology — *Dec. 2024*
- 2023 Innovation Practice Star – China Graduate Innovation Practice Series Competition — *Apr. 2024*
- First Prize in Huawei Cup National Graduate Mathematical Modeling Competition — *Dec. 2023*
- First-Class Academic Scholarship – Beijing Institute of Technology — *Oct. 2023*
- Beijing Outstanding Graduate — *Jun. 2022*
- BUCT Outstanding Graduation Thesis Award — *Jun. 2022*
- Li Wenyang and Yang Yan Social Support Scholarship (Top 0.1%) — *Nov. 2021*
- National Scholarship (Top 0.2%) x2 — *Oct. 2020, Oct. 2019*
- Outstanding Student × 2, Excellent Student × 2 — *Sep. 2019 - May. 2022*